CIVL 1100
Discovering Civil and Environmental Engineering

Introduction to Laboratory Sessions

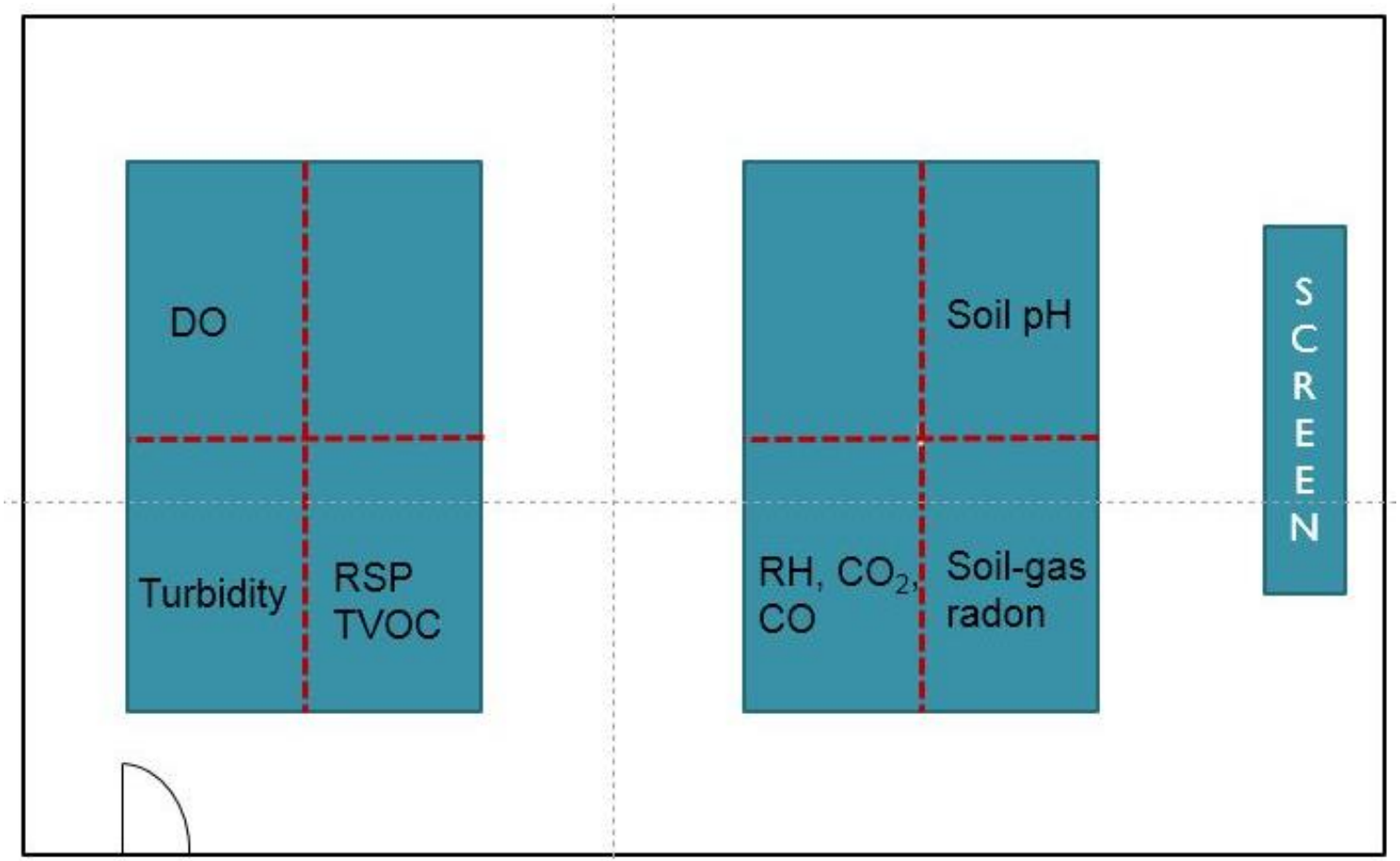
ENVIRONMENTAL ENGINEERING EXPERIMENT

Objectives of these Experiments

- Gain hands-on experience in conducting environmental experiments (soil, air, water).
- Have basic understanding of the tests related to soil, air and water during the lab session.

Floor plan

6 groups



RSP: respirable suspended particulates
TVOC: total volatile organic compounds
DO: dissolved oxygen
RH: relative humidity

Basic Rules

- Wear appropriate shoes and clothes.
  - Your shoes should be able to protect your feet.
- No eating and drinking during the lab session.
- Please use the lab equipment **carefully**.
- Please **follow the instructions** given by TAs, lab tutors and lab technician.

Introduction

- The lab session is composed of 3 parts:
  - Part 1: Soil
    - ✓ Experiment 1 (i): Soil pH
    - ✓ Experiment 1 (ii): Soil-gas radon
  - Part 2: Indoor Air Quality
    - ✓ Experiment 2 (i): Relative humidity, CO_2, CO
    - ✓ Experiment 2 (ii): RSP, TVOC
  - Part 3: Water
    - ✓ Experiment 3 (i): Dissolved oxygen
    - ✓ Experiment 3 (ii): Turbidity

RSP: respirable suspended particulates

TVOC: total volatile organic compounds

Part 1: Soil

- Experiment 1 (i): Measurement of Soil pH

Orion Model 250A pH Meter

Test Procedures

1. Stir the soil-water mixture for 30 seconds.
2. Let the mixture settle for 2 minutes.
3. Repeat steps 1-2 for the other two samples.
4. Rinse the electrode using the deionized water.
5. Wipe the electrode with Kimwipes tissue.
6. Insert the electrode into the upper layer of the sample (aqueous phase).
7. Record the reading of pH value after 30 seconds.
8. Repeat steps 4-7 for another two sets of measurement.

Part 1: Soil

- Experiment 1 (ii): Measurement of Soil-gas Radon Level

Durridge Model RAD 7 Electronic Radon Detector

Test Procedures

1. Ensure that the screen is on “>Test” mode.
2. Push “enter” button.
3. Push “→” button to select “>Test Print” mode.
4. Push “enter” button.
5. Record the reading of soil-gas radon printed from the wireless infrared printer.
6. Repeat steps 1-5 for another two sets of measurement.

Part 2: Indoor Air Quality

- Experiment 2 (i): Measurement of Relative Humidity, Carbon Dioxide (CO_2), and Carbon Monoxide (CO)

TSI IAQ-Calc™ Model 7545
IAQ Monitor

Test Procedures

1. Keep the probe surrounded by air.
2. Press "↵" button.
3. Record the readings for relative humidity, CO_2 and CO after 30 seconds.
4. Press "ESC" button.
5. Repeat steps 1-3 for another two sets of measurement.

Part 2: Indoor Air Quality

- Experiment 2 (iia): Measurement of Respirable Suspended Particulates (RSP)

TSI DustTrak™ II Model 8530 Desktop Aerosol Monitor

Test Procedures

1. Press the "Start" button on the screen.
2. Record the reading after 30 seconds.
3. Repeat step 2 for another two sets of measurement.

Part 2: Indoor Air Quality

- Experiment 2 (iib): Measurement of Total Volatile Organic Compounds (TVOC)

ppbRAE 3000
TVOC Monitor

Test Procedures

1. Ensure that the TVOC monitor is placed on the stand.
2. Record the reading after 60 seconds.
3. Repeat step 2 for another two sets of measurement.

Part 3: Water

- Experiment 3 (i): Measurement of Dissolved Oxygen

YSI Model 58

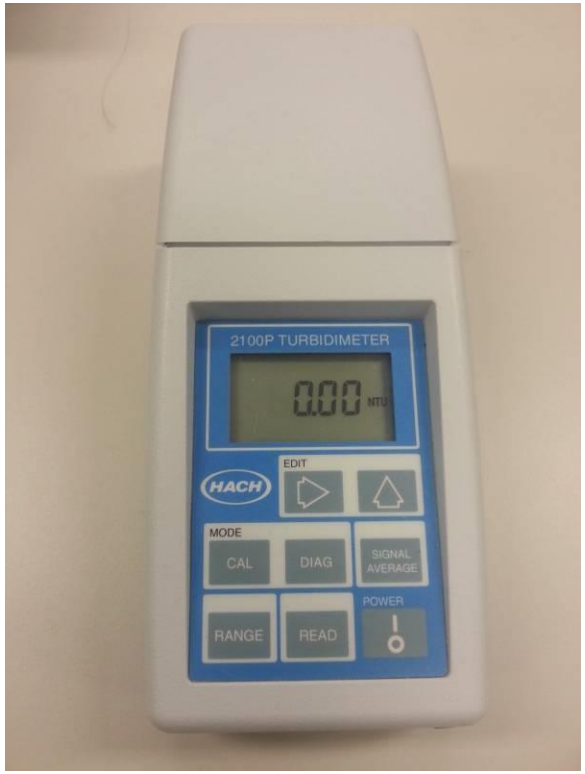
Dissolved Oxygen Meter

Test Procedures

1. Insert the probe into the wastewater sample.
2. Turn on the stirrer by pushing the red button on the probe.
3. Record the reading of dissolved oxygen after 30 seconds.
4. Turn off the stirrer.
5. Rinse the probe with deionized water.
6. Wipe the probe with Kimwipes tissue.
7. Repeat steps 1-6 for another two sets of measurement.

Part 3: Water

- Experiment 3 (ii): Measurement of Turbidity



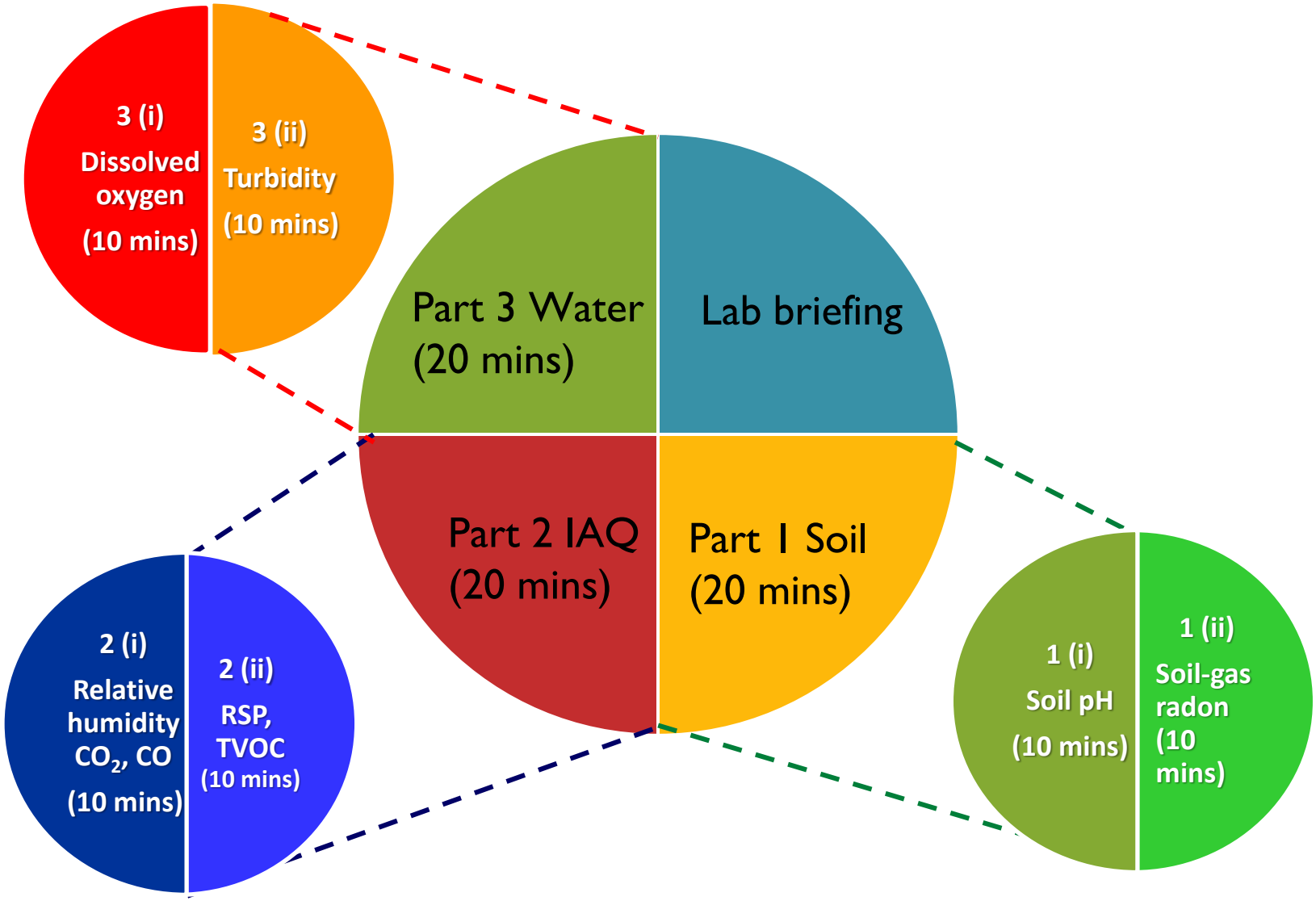
HACH 2100P Turbidimeter

Test Procedures

1. Fill the clean sample cell with tap water sample to the mark.
2. Cap the sample cell.
3. Wipe the sample cell with lint-free cloth.
4. Insert the sample cell into the cell compartment and close the cover.
5. Press the "READ" button.
6. Record the reading.
7. Discharge the tap water sample into the waste bottle.
8. Rinse the sample cell using deionized water.
9. Repeat steps 1-8 for another two sets of measurement.

Time Schedule

Total time = 1 hr 20 mins



Note: A stopwatch countdown will be shown on the TV screens.

Experiment Rotation

- Step 1: Sit next to the bench according to your group after getting into the lab
- Step 2: Move to the next part of experiment(s) (after 10 mins)*
- Step 3: Complete all the experiments in the sequence below:

$$1 \rightarrow 2 \rightarrow 3 \rightarrow 4 \rightarrow 5 \rightarrow 6$$

Lab Report on Data Recording Sheet and Question Answering

- WORD/PDF copies of the lab report template for data recording and question answering is now available in Canvas under “Laboratory”.
- You should record the data while doing experiments and answer all questions on the lab report.

Department of Civil and Environmental Engineering
The Hong Kong University of Science and Technology

CIVL 1100
Discovering Civil and Environmental Engineering

Lab Session for Environmental Engineering

Lab Report on Data Recording Sheet and Question Answering
(Spring 2024)

Full Name (in BLOCK letters)	Student ID

Due date of Submission: 2 April (For session LA1-1, 1-2, 2-1, and 2-2); 16 April (For session LA1-3, 1-4, 2-3, and 2-4)
Submit a PDF copy of your lab report to Canvas before 11:59 pm on the assigned due date (late submissions will not be accepted).

Lab Report Submission

- Please provide the full name and student ID of all the group members on the first page of the lab report.
- Each student in a group is required to submit the same report to CANVAS individually.
- The lab report is due One Week after your environmental lab session (late submissions will not be accepted).

Date of experiment	26 March
Report due date	2 Apr. before 11:59 pm

Date of experiment	9 April
Report due date	16 Apr. before 11:59 pm

If you have anything unclear, please feel free to message TAs of Part II:

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